

When is Bayesian optimization done?

An intern project brief, introductory report, and reproducible experiment specification

Prepared 25 September 2026. AI-written starting report to help the student and collaborators understand the main ideas. This package defines the study and provides a runnable function bank. The BoTorch optimizer, stopping implementations, and experimental results are student deliverables; the screen has not been run.

1. The question and the project boundary

The research question is: **For a given landscape and acquisition strategy, which stopping rule returns an acceptably good solution with the fewest evaluations, and how often does it stop prematurely?**

There is no universal, data-only test that establishes the global optimum of an arbitrary black-box function after finitely many observations. An unobserved narrow peak can always remain unless assumptions restrict what the function can do. A defensible answer therefore states a domain, an acceptable error, a model or regularity assumption, and an allowed risk.

The project should deliver an empirical map of when stopping rules work and fail. It should not promise a universally best rule or invent a new BO algorithm. The initial scope is scalar maximization, a bounded continuous input space, sequential evaluations, stationary GP surrogates, and two or three input dimensions. Use BoTorch directly so the student can see the model, policy, recommendation, and stopping decision separately.

Start by evaluating the continuous functions on a fixed, finite candidate catalogue and allowing all methods to select from that catalogue. This makes the optimum over the allowed actions exactly computable, even for random fields. A later, bounded follow-up uses continuous acquisition optimization. Finite-catalogue results must never be presented as certificates over the entire continuous surface.

First deliverable: review and refine this report so the student and collaborators share an understanding of the main ideas, assumptions, and limitations. Check the sources and note questions that need clearer explanations or examples before implementing the screen. Suggested prerequisites are Python/NumPy, probability distributions, conditional probability, and elementary matrix algebra. Deriving new convergence theorems is outside scope.

2. Four objects that must remain separate

- The **objective function** $f(x)$ represents the scientific quantity being maximized.
- The **acquisition function or policy** decides where to evaluate next.
- The **recommendation rule** selects the solution returned to the user.
- The **stopping rule** decides whether enough evidence or progress has accumulated.

An acquisition function is an optimization target inside BO, not the experimental objective itself. A solver reporting convergence while maximizing an acquisition function also does not mean the outer BO experiment is done.

If “optimization targets” means scientific goals, five distinct possibilities are: finding one best setting, maximizing cumulative reward during experimentation, finding settings above a threshold, learning a whole response surface, and identifying a multi-objective Pareto set. These require different success definitions. This project studies the first. In particular, finding one maximum is not the same as mapping

all peaks or accurately reconstructing a surface.

3. BO in enough depth to explain it

Observe $y_i = f(x_i) + e_i$, where e_i is measurement noise. A Gaussian process places a joint Gaussian distribution over function values. Its mean expresses the current prediction; its covariance expresses how information at one location changes beliefs elsewhere. A length scale controls how quickly correlation decays with separation. Separate length scales per coordinate are called automatic relevance determination, or ARD.

For a zero-mean GP with fixed covariance k and independent Gaussian noise covariance Σ , the posterior at a new location is:

$$\begin{aligned}\mu_t(x) &= k(x, X) [K(X, X) + \Sigma]^{-1} y \\ s_t^2(x) &= k(x, x) - k(x, X) [K(X, X) + \Sigma]^{-1} k(X, x)\end{aligned}$$

Implement this with linear solves, not a matrix inverse. The latent standard deviation s_t concerns uncertainty about f . The uncertainty of a new noisy observation additionally includes its noise variance. Repeated measurements can reduce uncertainty about the latent mean even when individual observations remain noisy.

The loop is: obtain an initial design; fit the GP; construct a policy from its posterior; choose a new location; observe it; update; evaluate the stopping condition. Exploration visits uncertain regions that might be better. Exploitation visits regions already predicted to be good. Neither behavior alone solves all landscapes.

GP uncertainty is conditional on assumptions. A fitted length scale that is too long can smooth away a narrow peak and give false confidence. Optimizing hyperparameters and then treating them as known omits hyperparameter uncertainty. A well-fitted mean curve is therefore insufficient evidence that a stopping probability is calibrated.

Let $x_{\hat{t}}$ be the recommended point, f_{star} the best allowed value, and r_t the simple regret:

$$\begin{aligned}f_{\text{star}} &= \max_{x \text{ in domain}} f(x) \\ r_t &= f_{\text{star}} - f(x_{\hat{t}})\end{aligned}$$

The central success event is $r_t \leq \epsilon$. For the core noiseless study, recommend the best evaluated point. In the noisy extension, recommend the evaluated point with highest current posterior mean. Do not report the highest noisy observation as if it were the true best value. Keep recommendation conventions identical across policies and stopping rules.

Cumulative regret sums losses over all evaluations. It matters when each experimental action itself earns a reward, but it is not this project's primary objective. Coordinate error is also secondary: two widely separated settings can have equally good values.

4. Five acquisition strategies

These are five established, representative strategies, not a bibliometrically verified popularity ranking. EI, PI, and UCB are classical choices; Thompson sampling provides a randomized policy; MES represents information-based acquisition. Knowledge gradient is a valuable reading topic but is excluded from the first screen to keep implementation bounded.

ID	Strategy and target	Expected behavior to test	BoTorch implementation route
A1	Expected improvement (EI): expected positive gain over the incumbent	Balances gain and uncertainty; may miss a narrow region if the model becomes overconfident	LogExpectedImprovement for the noiseless $q=1$ study
A2	Probability of improvement (PI): probability of beating incumbent plus margin ξ	Can chase small, likely improvements; sensitivity to the improvement margin is instructive	LogProbabilityOfImprovement with $\text{best_f} = \text{incumbent} + \xi$
A3	Upper confidence bound (UCB): posterior mean plus κ times standard deviation	Explicit exploration control; insufficient κ can become greedy	UpperConfidenceBound, with $\text{beta} = \kappa^2$
A4	Thompson sampling (TS): maximize one jointly sampled posterior function	Randomizes among plausible optima; posterior correlation must be preserved	MaxPosteriorSampling over the shared candidate catalogue
A5	Max-value entropy search (MES): information about the unknown maximum value	May sample points valuable for resolving the optimum rather than immediate improvement	$q\text{MaxValueEntropy}$, $q=1$, with the shared candidate catalogue

For a Gaussian latent posterior, define $z = (\mu - b - \xi)/s$. With ϕ and Φ denoting the standard normal density and distribution function:

$$\begin{aligned} \text{PI}(x) &= \Phi(z) \\ \text{EI}(x) &= (\mu(x) - b - \xi) \Phi(z) + s(x) \phi(z) \\ \text{UCB}(x) &= \mu(x) + \kappa s(x) \end{aligned}$$

Handle $s = 0$ by the limiting expression. The initial EI policy uses $\xi = 0$; the PI policy uses $\xi = 0.01$ in the function bank's declared output units. Set $\kappa = 2$ for the pilot. These are predeclared study settings, not universal recommendations. Treat modest sensitivity checks as follow-up work, not a new Cartesian product in the pilot.

Use log-space EI/PI implementations to avoid numerical underflow in acquisition optimization. A small raw EI caused by floating-point failure must not trigger scientific convergence. Compare $\log(\text{EI})$ to $\log(\text{threshold})$, not to a threshold in raw objective units. Numerical improvements to EI are documented by Ament et al. [R3]. BoTorch API mappings are in [R2]; the TS and MES tutorials are [R4-R5].

MES targets mutual information between a possible observation and f_{star} . It is not the same as simply sampling where posterior variance is largest. Read Wang and Jegelka [R6] for its derivation. Its acquisition values have information units, so a universal numerical threshold shared with EI or UCB would be meaningless.

5. A function bank organized by failure mode

All implementations accept arrays with final dimension d and inputs in $[0, 1]^d$. Standard functions are mapped into their stated native bounds, negated when originally defined as minimization problems, and scaled by fixed, documented constants. The supplied source implements these choices. Most functions extend directly to $d > 3$; the named fixed-dimensional anchors do not.

Family	Property isolated	What failure would teach us
Quadratic peak	One smooth global maximum	Basic implementation and efficient stopping on an easy case
Rotated ellipsoid	One maximum, condition number 100, coupled coordinates	Sensitivity to anisotropy and poorly aligned length scales

Family	Property isolated	What failure would teach us
Rippled dominant peak	Broad trend plus smooth short-scale oscillations; known maximum	Premature stopping among local peaks; distinguishes short-scale variation from observation noise
Competing Gaussian peaks	Several smooth, separated peaks with similar amplitudes	Need to rule out credible competitors, rather than stabilize within one basin
Rosenbrock	Narrow curved valley before negation, strong coordinate coupling	Slow progress can reflect difficult geometry rather than completion
Ackley	Many local features with a dominant optimum and relatively flat outer region	Exploration across an initially uninformative landscape
Rastrigin	Repeated competing local structure	Sensitivity to multimodality and periodicity
Hidden narrow peak	Broad attractive feature plus a narrow, better remote feature	Confident but wrong stopping after the best region is missed
GRF, long scale	RBF-like covariance, length scale 0.25	Broad stochastic landscapes without hand-selected extrema
GRF, short scale	RBF-like covariance, length scale 0.06	Many small-scale features; covariance mismatch
GRF, rougher covariance	Matern 3/2 spectral approximation, length scale 0.12	Misspecification when the fitted surrogate assumes smoother variation
Branin anchor	Standard 2D test with multiple equivalent global minima	Coordinate convergence is not necessary for value convergence
Hartmann-3 anchor	Standard multimodal 3D test	A familiar benchmark for implementation comparisons

The first eleven families appear in both 2D and 3D: 22 cases. Add Branin in 2D and Hartmann in 3D: **24 cases**. Standard-function background is available from the SFU test-function collection [R11]. The custom functions and parameter choices are proposed experimental design, not claims from that collection.

A differentiable function with rapid oscillations is not mathematically nonsmooth. The report and figures must distinguish short length scales, high curvature, nonstationarity, and stochastic observation noise. The starter's finite spectral Matern approximation is still smooth at sufficiently fine scales; it is not an exact nondifferentiable Matern sample path.

Gaussian random fields: freeze the function

Generate one latent field per field seed and keep it fixed throughout every optimization run. Add observation noise separately. Resampling the field at each query would change the optimization problem.

The supplied generator uses fixed frequencies and Gaussian coefficients:

$$f(x) = (1 / \sqrt{M}) \sum_m [a_m \cos(w_m \cdot x) + b_m \sin(w_m \cdot x)]$$

a_m, b_m independently distributed as $N(0, 1)$

Conditional on the frequencies, this is an exact finite-rank Gaussian field, with covariance equal to the average of $\cos(w_m \cdot (x - x_{\text{prime}}))$. Its covariance approximates RBF or Matern covariance through spectral sampling. It is not an exact draw from the corresponding infinite-rank kernel. $M = 512$ is a starter setting; verify resolution sensitivity before interpreting fine-scale behavior.

For the pilot, use one predeclared field realization per covariance regime and dimension, and five optimizer seeds on each. This measures optimizer variability on fixed fields. In confirmation, add at least five field realizations for the regimes of interest. Field seed and optimizer seed are distinct experimental factors.

An essential small validation fixture should also draw an exact multivariate Gaussian on the finite catalogue, fit the same known covariance with fixed parameters, and assess stopping there. This tests implementation under matched assumptions before studying fitted, potentially wrong surrogates.

6. Five stopping approaches

Define “done” operationally as a tolerable loss in objective value. For pilot scoring use $\epsilon = 0.02$ in the supplied function units, and also report curves for $\epsilon = 0.01$ and 0.05 . Fixed scaling makes this reproducible, but equal numeric tolerances are not automatically equal scientific difficulty. Report each case separately and inspect the tolerance against its peak-to-peak differences before freezing the confirmatory study.

All early stopping begins only after $n_{\min} = n_{\text{initial}} + 10$ observations. Check every five subsequent observations and at the final budget. A check occurs after incorporating the latest observation and refitting. The stopping code only sees the information available at that prefix.

ID	Approach	Operational rule	Principal limitation
S1	Fixed budget	Stop at a predeclared evaluation limit	Controls resources, makes no claim about optimality
S2	Incumbent stagnation	Best evaluated value improves by at most η over the last w observations	May be trapped, underexploring, or crossing a difficult region
S3	Negligible expected improvement	Maximum EI across the allowed domain is below τ_{EI}	Conditional on the surrogate; small EI is not a probability of global success
S4	Global confidence gap	Best plausible value minus lower bound at the returned point is $\leq \epsilon$	Requires valid simultaneous bounds; plug-in GP intervals need not provide them
S5	Posterior probability of ϵ -optimality	Posterior probability that returned point has regret $\leq \epsilon$ exceeds the specified level	Model error and numerical probability estimation can produce false confidence

S2 and S3 are useful heuristic baselines. S4 and S5 are attempts to relate stopping to solution quality under assumptions. No evidence supports calling these an exact ranking of the five most common stopping rules. Budget, convergence, and acquisition thresholds are established baselines; regret-based stopping is an active research topic [R7-R9].

S1: budget

Use $B = 100$ total observations in 2D and $B = 150$ in 3D, including the initial design. Also score smaller fixed-budget prefixes at $n_{\min}$ and at approximately 50% and 75% of B . Every other stopping method has the same B as a fallback. Distinguish “criterion fired” from “budget exhausted without a criterion firing.”

S2: stagnation

In the noiseless core, $b_t = \max_{i \leq t} y_i$. Stop if $b_t - b_{(t-w)} \leq \eta$, with $w = 10$ and $\eta = 0.002$. This definition is monotone and reproducible. A single long flat interval does not rule out an unseen peak.

For a noisy follow-up, use a separately labeled denoised version: re-evaluate the best posterior mean among all evaluated points and among the prefix excluding the last w observations, using the same current posterior for both. This avoids confusing changing GP fits with genuine incumbent improvement, but remains a heuristic.

S3: small expected improvement

Compute a separate EI diagnostic regardless of which policy generated the trajectory. Stop when $\max_x \text{EI}_t(x) \leq 0.001$ in the declared function units. This retains a common stopping rule for all five policies; applying the same raw threshold to their own acquisitions would compare incompatible quantities. The maximization is exact over the finite catalogue, up to numerical evaluation.

For the noisy extension, replace EI with a noise-aware latent improvement diagnostic using `qLogNoisyExpectedImprovement` and an evaluated-point baseline. Label that as the noisy adaptation rather than pretending ordinary EI with the largest noisy y is valid. Always account for the diagnostic's own computation time.

S4: a global confidence gap

Define upper and lower envelopes of the latent function, $U_t(x) = \mu_t(x) + k_t s_t(x)$ and $L_t(x) = \mu_t(x) - k_t s_t(x)$. For the common recommendation x_{hat_t} , compute:

```
g_t = max_{x in Q} U_t(x) - L_t(x_hat_t)
stop if g_t <= epsilon
```

If all true function values are simultaneously inside the envelopes, then $f_{\text{star}} \leq \max U$ and $f(x_{\text{hat}_t}) \geq L(x_{\text{hat}_t})$, so $r_t \leq g_t$. The lower bound must be at the point actually returned; $\max U - \max L$ would certify a different recommendation unless those points coincide.

For a known, correctly specified GP on a fixed catalogue Q of size M and J predeclared checks, a conservative Gaussian-tail union-bound construction is $k = \sqrt{2 \log(2 M J / \delta)}$, with $\delta = 0.05$. This is a useful matched-model validation control. It is not a blanket theorem for fitted hyperparameters or arbitrary fixed benchmark functions. For the main fitted-GP study, retain the same formula but label it a model-based bound and measure coverage empirically. Ordinary pointwise 95% intervals do not establish simultaneous 95% coverage over locations and repeated checks.

S5: probability of acceptable regret

Wilson [R7] studies stopping when a point is epsilon-optimal with high posterior probability. The project uses that criterion, with a deliberately simple finite-catalogue implementation:

```
p_t = P(max_{x in Q} f(x) - f(x_hat_t) <= epsilon | observations)
```

Draw joint latent posterior samples over Q , including the recommendation. In each draw, subtract the sampled value at the recommendation from the maximum of that same draw. Do not sample the maximum and the recommended value independently.

Use 512 independent Gaussian posterior draws for the pilot. Count the fraction satisfying the event and use a one-sided binomial lower confidence limit. With a nominal total model-based error allowance of 0.05, allocate 0.04 to posterior regret and 0.01 to Monte Carlo decision error across the run. Stop only if the lower limit exceeds 0.96, with per-check Monte Carlo error $0.01/J$. Fixed draws and checks avoid repeated unchecked Monte Carlo peeking. Binomial limits require independent draws; do not substitute quasi-Monte Carlo samples without revising that error calculation.

These choices control sampling error under exact posterior sampling, not model misspecification. Approximate low-rank samplers introduce another error source that must be labeled and checked. This implementation is a study adaptation, not a claim to reproduce every detail or guarantee of Wilson's algorithm.

Other literature the student should recognize

Makarova et al. [R8] relate useful remaining optimization to the statistical error of a hyperparameter objective. That motivates a noise-aware interpretation of tolerance; it does not mean the standard deviation of a single noisy measurement is a universal stopping floor.

Ishibashi et al. [R9] propose stopping based on the change in expected minimum simple regret. The student should explain how a change in expected regret differs from a bound on its absolute level; full reproduction is outside the initial five-rule screen.

Xie et al. [R10] explicitly study evaluation costs and stopping. Include this in the reading overview, but keep variable costs and their acquisition rules out of the first implementation. “Expected improvement smaller than cost” requires a conversion between objective gain and cost; it is not meaningful when the quantities have unrelated units.

7. Frozen optimization protocol

The following numerical settings are proposed study choices, not literature recommendations.

1. Use a fixed scrambled Sobol catalogue Q with 2,048 points per dimension setting. Save its coordinates and seed. Use the same Q across functions and policies of a given dimension. All initial and subsequent queries must come from Q in the first screen.
2. Use $n_{\text{initial}} = 5d$ distinct catalogue points, sampled without replacement. Pair their indices across policies for each optimizer seed. Candidate construction and initial-design randomness are separate.
3. Fit a SingleTaskGP with an explicitly supplied scaled ARD Matern-5/2 kernel. Do not rely on library-default kernel choices. Use double precision, inputs already in $[0,1]^d$, and an outcome standardization transform. Supply a small, fixed numerical noise variance for the noiseless stage and record its value and any library clamping. It is not physical observation noise.
4. Fit kernel hyperparameters by marginal-likelihood optimization after every observation, with fixed initialization/restart policy and logged failures. Preserve priors, bounds, and fitted values in the run record. Do not silently change settings on difficult functions.
5. Use $q = 1$. Optimize EI, PI, UCB and MES by evaluating them over Q ; TS samples jointly over Q . Remove previously evaluated points from the selectable actions in the noiseless stage, but keep the full Q for global stopping diagnostics. Compute acquisition values in manageable chunks without breaking joint TS sampling.
6. Apply the common recommendation rule. At scheduled checks, compute all five stopping rules from that prefix. Save their diagnostics and first trigger times, but continue the policy to B .
7. Give separate random generators to initialization, policy sampling, model restarts, observation noise, posterior stopping draws, and field generation. Adding a stopping rule must not change the future acquisition trajectory.
8. Keep the complete trajectory. Truth used for scoring is held in an evaluator that cannot be accessed by the optimizer or stopper. Evaluation of all Q values is offline benchmark construction, not free information available to BO.
9. Add a uniform random-search baseline on Q with the same evaluation budget. It is a reference optimizer outside the five BO policies. Do not expand this into a competition among libraries.
10. Record package versions, hardware, configuration, data hashes, warnings, and random seeds. Lock the environment after the first working BoTorch example. Pin the versions actually tested rather than claiming this starter already supplies a verified BO environment.

The noisy extension uses $y = f + N(0, 0.02^2)$, with the variance given to the model. Allow repeat evaluations there and count each one. EI becomes noise-aware EI; PI uses a documented posterior-mean

incumbent heuristic. Do not pool noiseless and noisy variants as if they were identical algorithms. Unknown, heteroscedastic, or heavy-tailed noise is later work.

8. Full screen without duplicated trajectories

For one noise setting, the pilot comprises 24 cases x 5 acquisition policies x 5 optimizer seeds = **600 full trajectories**. Applying five stopping rules gives **3,000 case-policy-stopper-seed outcomes**. These are correlated outcomes, not 3,000 independent runs. Fixed-threshold sensitivity scores can be computed from the same stored diagnostics.

There are 12 cases in 2D and 12 in 3D. At the proposed budgets, the main BO screen uses at most 75,000 objective evaluations, including initial designs. The random-search reference adds 120 trajectories and 15,000 evaluations. Synthetic function evaluation is cheap; GP fitting, joint sampling, and stopping diagnostics may dominate the runtime. Profile a small run before launching the screen.

A second noise setting doubles the main totals to 1,200 trajectories and 6,000 scored outcomes. It is a gated second stage, not a prerequisite for the first useful result.

Shadow evaluation is valid because each rule only decides whether to terminate. It must not change sampling, request extra confirmation observations, or use future observations to recompute an earlier decision. Any stopping policy that actively requests validation measurements becomes a distinct experimental policy and needs a real separate run.

Minimum launch ladder: one quadratic example; then three contrasting functions x two policies x two seeds with full diagnostics; then the complete five-seed screen. Complete the full planned matrix rather than showing only favorable examples. Increase seeds on selected comparisons after the pilot identifies remaining questions.

9. How the study judges success

Measure	Definition and interpretation
Final simple regret	Exact catalogue maximum minus true latent value of the point returned at stopping
False-stop rate among triggers	Fraction of early criterion triggers with regret above epsilon; report numerator and denominator
Unconditional premature-stop frequency	Fraction of all runs that trigger early and return regret above epsilon
Success at termination	Fraction returning regret $\leq$ epsilon, including budget terminations
Evaluation cost	Total observations at termination, including initialization and any repeats
Compute cost	GP fitting, acquisition selection, and each stopping diagnostic timed separately
Non-trigger fraction	Fraction reaching B without the stopping condition firing
Regret-versus-cost frontier	Compare cost at comparable reliability instead of announcing a winner from average cost alone
Late-stop overhead	Extra observations after first epsilon-optimal recommendation; interpret with care when noisy recommendations can later worsen
Model diagnostics	Predictive coverage/error on a fixed audit set, length scales, and uncertainty in missed regions

Truth and audit locations are evaluator-only. Do not use their outcomes to adjust a stopping threshold on the test cases. Calibration/tuning uses separate seeds or field realizations, and confirmation uses held-out ones. Benchmark truth can be used for scoring and offline figure production only.

Report every individual seed for the five-run pilot, plus median and range. Avoid elaborate significance claims. Zero premature stops in five independent runs still permits a one-sided 95% binomial upper bound of about 45% for that failure probability. Roughly 59 independent zero-failure runs are needed to put that upper bound below 5% for one specified condition. Pooling unrelated landscapes does not establish a 5% bound on each landscape.

Use 20-30 seeds for promising or problematic combinations to estimate variability more usefully; use a sample-size calculation if the goal becomes a reliability claim. For multiple field realizations, preserve the hierarchy: compare policies within the same field and optimizer seed, then summarize across fields. Do not treat every stopping-rule score or every time point as an independent replicate.

Required plots: true landscape with sampled locations and stop markers; regret versus evaluations; false-stop frequency versus median evaluations; distribution of stopping times; and a case-by-policy-by-stopper table. Show a successful case and a failure case for every major conclusion. For 3D, use several fixed slices and objective-versus-time plots rather than one opaque surface rendering.

10. From catalogue results to continuous optimization

The catalogue is a controlled first experiment, not a substitute for the continuous problem. In 3D, a finite set can completely miss a narrow peak. Quantify that error on functions with known continuous optima and with dense/offline searches on the other functions.

After the pilot, select two acquisition strategies and two stopping rules, based on a written scientific question rather than favorable rankings. Re-run the relevant cases with continuous acquisition optimization through BoTorch's `optimize_acqf` (TS requires a continuous posterior-path approach or a clearly labeled candidate approximation). Freeze multistart settings and inspect numerical optimization failures.

For continuous functions with known extrema, use those for exact regret. For Gaussian mixtures and GRFs, a dense search plus multistart local optimization supplies a numerical reference, not a proof of the global optimum. The best value found is a lower bound on the true maximum, so the resulting estimated regret can understate true regret. Mark success classifications as provisional if reference uncertainty is material. Neither max U nor sampled-function maxima on a coarse grid are valid global continuous upper bounds.

Perform catalogue-size and spectral-resolution sensitivity checks on these selected cases. Expansion to $d = 5$ or 10 comes later; a fixed number of catalogue points becomes much sparser as dimension grows. An N-dimensional function API removes coding barriers but does not remove the curse of dimensionality or exact-GP cost.

11. Reading sequence and shared understanding

Use this AI-written draft as a starting point for shared learning. The student and collaborators should review the key ideas, check the sources, improve unclear explanations, and record open questions. A source-to-claim table can help connect the methods to the literature.

1. Read selected introductory/modeling/policy sections of Garnett [R1]. Explain conditioning, posterior mean/variance, simple regret, and exploration versus exploitation using a one-dimensional drawing.
2. Work through the official BoTorch loop [R2a]. Identify what gets fitted, what gets optimized, what a function evaluation costs, and why $q=1$ simplifies the study. Run one complete toy loop before adding abstractions.

3. Read Ament et al. [R3], then explain why logEI is a numerical change rather than a new scientific objective. Inspect the TS and MES examples [R4-R6].

4. Read the introductory method and experiments of Wilson [R7]. Explain the distinction between posterior probability and empirical reliability under misspecification.

5. Read Makarova [R8] and Ishibashi [R9] for contrasting stopping targets. Skim Xie [R10] for the role of costs. Full proofs and reproducing all their algorithms are not required.

Aim for a concise overview of approximately 6-8 pages excluding references and appendices, written so the student and other collaborators can follow the main points. For each method it must contain: one-paragraph intuition, defining equation, assumptions, implementation link, tunable quantities, one likely failure case, and the metric that would reveal that failure. AI assistance is allowed, but every citation and implemented equation must be checked against the source.

Questions to guide reading and discussion:

- What exactly is being optimized at the outer and inner levels?
- Why can a GP be confidently wrong about a hidden peak?
- Why is lack of recent progress not evidence of global optimality?
- What is the difference between latent uncertainty and observation noise?
- Why does small EI not establish a 95% probability of success?
- Why are joint samples required for Thompson sampling and posterior regret?
- Derive the global confidence-gap inequality in two lines.
- Why can the same trajectory evaluate multiple stopping rules fairly?
- What changes if the stopper requests additional measurements?
- What do five seeds establish, and what do they not establish?
- What is exact about a catalogue optimum, and what is not continuous about it?
- How can an approximate reference optimum hide a failure?

Use these questions to identify topics that need clearer explanations or worked examples. Reproducing a simple run from a clean checkout helps connect the ideas to practice.

12. Software overview without scope expansion

Tool	Role the student should understand	Role in this project
BoTorch	Research-level models, acquisitions, posterior sampling and acquisition optimization	Required implementation engine
GPyTorch	Gaussian-process modeling underneath common BoTorch GP models	Understand kernel/likelihood configuration; no custom GP engine
Ax	Experiment management and orchestration around adaptive optimization	Read the overview; not a second implementation
Optuna	Hyperparameter study management, sampling, pruning and termination tooling	Compare concepts; distinguish stopping a trial from stopping the whole study
SciPy	Conventional local/global optimization and numerical/statistical routines	Reference searches, Sobol designs and statistical calculations

Official documentation is linked in [R12-R14]. As inspected on 25 September 2026, Optuna's in-package terminator documentation marks that module deprecated, with functionality planned for OptunaHub. Do not design this project around an unchecked old Optuna API. Read algorithm papers for the concepts and use BoTorch for the main study.

13. GitHub exercise and milestones

Use one repository, small pull requests, saved configurations, and executable examples. The proposed issues are listed in docs/project_issues.md. Do not write a general optimization platform.

Milestone	Deliverable	Completion check
1: shared understanding	Clear overview, checked sources, and notes on open questions	Main ideas, assumptions, and limitations are clear to the student and collaborators
2: one complete run	One BoTorch loop and one reproducible plot	Fresh checkout reproduces it; budget and recommendation clear
3: function bank	24 cases, 2D gallery, 3D slices, separate field seeds	Known-value tests pass; repeated field queries agree
4: stopping implementations	Five rules with logged diagnostics	No future-data access; matched-model and adversarial cases checked
5: pilot screen	Complete five-seed matrix and summary tables	All planned cells accounted for, including failures and non-triggers
6: focused confirmation	Extra seeds/fields and selected continuous cases	Thresholds frozen; uncertainty and discretization limitations reported
7: public-facing package	README, examples, report, small result files	Another person reproduces one figure from documented commands

An illustrative schedule is two weeks for reading and the first loop, one week for benchmarks, one week for stopping rules, then two weeks for screening, confirmation, and documentation. Adjust this to the intern's preparation and compute profile. Use the milestones to guide progress rather than relying on a

fixed calendar.

Out of scope until the core result is reviewed: new acquisition functions, neural surrogates, large GPs, multi-objective BO, constraints, asynchronous batches, multi-fidelity optimization, online field regeneration, and infrastructure/dashboard development.

14. Annotated sources

The numerical protocol, custom benchmark design, seed counts, and milestone choices in this document are proposed study design. Sources below support background and algorithmic families; they do not endorse those numerical settings. Official pages were checked on 25 September 2026.

[R1] Roman Garnett (2023), Bayesian Optimization. Primary introductory text; select relevant sections rather than assigning the entire book. <https://bayesoptbook.com/>

[R2] BoTorch acquisition API. Verify signatures against the version used in the repository. <https://botorch.readthedocs.io/en/stable/acquisition.html>

[R2a] BoTorch, Closed-loop batch, constrained BO with qLogEI and qLogNEI. Use as a loop reference, while simplifying to this project's unconstrained $q=1$ setting. https://botorch.org/docs/tutorials/closed_loop_botorch_only

[R3] Ament et al. (2023), Unexpected Improvements to Expected Improvement for Bayesian Optimization, NeurIPS. Numerical issues and log-space acquisition calculations. <https://arxiv.org/abs/2310.20708>

[R4] BoTorch, Bayesian optimization with large-scale Thompson sampling. Reference for correlated posterior sampling; large-scale variants are not required here. https://botorch.org/docs/tutorials/thompson_sampling

[R5] BoTorch, The max-value entropy search acquisition function. Practical MES implementation. https://botorch.org/docs/tutorials/max_value_entropy

[R6] Wang and Jegelka (2017), Max-value Entropy Search for Efficient Bayesian Optimization, ICML. Primary MES formulation. <https://proceedings.mlr.press/v70/wang17e.html>

[R7] James Wilson (2024), Stopping Bayesian Optimization with Probabilistic Regret Bounds, NeurIPS. Posterior epsilon-optimality and its computation; model mismatch remains material. https://proceedings.neurips.cc/paper_files/paper/2024/hash/b204de7078301292a8876a762eed3dcb-Abstract-Conference.html

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Appendix: benchmark landscape gallery

These are the supplied latent 2D functions, not BO results. Each panel has its own color scale. The random fields are fixed realizations.

